

DAY-05(24.07.2026) and DAY-06(25.07.2026)

OSI MODEL:

conceptual framework

ISO - International Organization for Standardization

1. Application layer
2. Presentation layer
3. Session layer
4. Transport layer
5. Network layer
6. Data link layer
7. Physical layer

APPLICATION LAYER:

- direct interface b/w user and network
- data → sends and receives the information
- user interface → window for network based applications to access network services
- **Protocols:** HTTP HTTPS DNS FTP SMTP
- NVT -[Network Virtual Terminal] → user to remote system(server) → ex: we type commands on any terminal but the server understands it
- FTAM - [File Transfer, Access, and Management] → access the files in different OS
- Directory Services → like learning platform → that authenticate and provides specific resources

PRESENTATION LAYER [Translation layer]:

- data → formatted, secured, compressed for receiving it correctly (computer to computer) → acts as a translator b/w user OS and other user OS . → machines understand each other's format
- like encode and decode the data respective of the computer
- encryption (ciphertext) and decryption → protocols (TLS/SSL)
- Compression : Reduces bit no for transmission → increases speed

SESSION LAYER [Dialogue Manager]:

- opening, closing, security of communication
- Session Lifecycle - opens a session, keeps it active, terminates the session
- Authentication and Security - verifies the users and secure the transmission
- Synchronization - checkpoints in data stream → communication resume from where it left off after failure instead of starting from begin (ex: file downloading)
- Dialog control - how two applications talk with each other
- half duplex : Only one data can send data at a time (walkie-talkie)
Full-duplex: Both sides can send and receive data
SIMULTANEOUSLY (Video Calls).

TRANSPORT LAYER:

- end - end delivery messages
- Protocol → TCP, UDP, SCTP
- Data → segments (TCP) / Datagrams (UDP)
- Service point addressing → uses port numbers → data delivered to specific process/ application and not the device (ex: bandit)
- Segmentation and Reassembly - splitting of messages and reassembling at correct destination [every app has port number] ex: web traffic (http(80) https(443)), email(25), file(21)
- Protocols - TCP(reliable)[connection-oriented → handshake(connection establishment and checks errors], UDP(fast)[connectionless no guarantee on delivery]

NETWORK LAYER:

- data → b/w host across different networks → handles path finding and logical addressing
- segments → encapsulated into packets
- Logical addressing → IP address → to packet
- Routing → most efficient path source to destination
- Hardware devices → routers, switches
- Protocol → IP, ICMP, IGMP, OSPF

DATA LINK LAYER (DLL):

- bridge b/w physical hardware and logical network
- ensures that data transfer is error free → physical medium
- Protocol → Ethernet, PPP, PPTP

1.Data Packaging(Framing) :

- packets → frames
- start stop bits → where one unit ends and next begins

2.Addressing Hardware:

- Physical Addressing → MAC(Media Access Control)
- Encapsulates the sender's and receiver's MAC addresses into the frame header

(frame header:

- **Destination MAC address** → tells the network which device should receive the frame.
- **Source MAC address** → identifies the sender device.)

- Switches, Bridges

3.Sublayers:

- Logical Link Control → manages flow and error control
- Media Access Control → handles hardware addressing and how devices physically access the transmission medium

4.Error Control:

- retransmits damages / lost frames

5.Flow Control:

- synchronizes the data rate → a fast sender and a slower receiver → to prevent data "bottlenecks" or loss.

6.Access Control:

- MAC sublayer → right to transmit at given time → prevents collisions

PHYSICAL LAYER:

frames → bits

bridge for actual physical connections b/w devices

Protocol → USB, SONET/SDH

- **Core responsibility:** Handles hardware-level transmission of bits (0s and 1s).
- **Essential hardware:** Hubs, repeaters, modems, and cables operate here.
- **Bit synchronization:** Keeps sender and receiver in sync using clock signals.
- **Bit rate control:** Defines transmission speed in bits per second.
- **Physical topologies:** Structures like bus, star, and mesh layouts.
- **Transmission modes:** Simplex (one-way), half-duplex (two-way, one at a time), full-duplex (two-way simultaneously).

▼ SECURE SOCKET LAYER:

Message Authentication Code

- cryptographic protocol → provides secure communication
- encrypted link → web server and web browser
- uses encryption → secure transmission of data
- Encryption → protects data
- Authentication → verifies and identifies the server to ensure that data is sent to the correct destination
- Data Integrity → ensures that data not altered during transmission
- Fortezza Algorithm → strong encryption → early SSL (military and government)

▼ TRANSPORT LAYER SECURITY:

Hashed Message Authentication Code protocol

successor to SSL → improves security and efficiency

offers → strong security, better performance. protection against modern cyber threats

Enhanced Encryption → strong encryption > SSL

Forward Secrecy →

- Every time your browser connects to a website, a **temporary session key** is created to encrypt that specific communication.
- The server also has a **long-term private key** (stored on the server).
- With **Forward Secrecy**, the temporary session key is generated independently of the server's private key.
- **Server private key**: A long-term key used to prove the server's identity.
- **Session key**: A temporary key used to encrypt one specific connection.

Improved Performance → better and efficient with optimized algorithm

host-to-host layers →

Transport, session, presentation, application